

Notes: Bernstein (JF, 2021)

Negative Home Equity and Household Labor Supply

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1 Big picture

- **Question.** Does negative home equity reduce household labor supply?
- **Setting.** The paper uses U.S. household-level administrative data from a major financial institution, 2010 to 2014, with detailed information on mortgages, liabilities, assets, and deposit inflows.
- **Why this is hard.** Home equity and labor income move together for many reasons that are not causal: local labor demand, borrower selection, house-price shocks, prepayment behavior, and household-specific wealth choices.
- **Main idea.** Compare households exposed to the same local house-price path but who bought at different times in the same area. Timing of purchase changes current equity positions even when households face the same later local shocks.
- **Main contribution.** The paper provides the first causal evidence that negative home equity reduces labor supply, rather than only affecting labor demand or consumption.
- **Core empirical design.** Construct a “synthetic” loan-to-value ratio using only origination timing, origination cohort median leverage, and local house-price changes. Use whether synthetic LTV exceeds 100% as an instrument for whether the household is actually underwater.
- **Main result.** Instrumented negative home equity reduces household labor income by about 2.2% to 5.6%, with baseline estimates around \$298 per month or 3.47%.
- **Main nonlinearity.** The drop in income appears only when households get close to or cross into negative equity. Large house-price movements that do *not* push households into negative equity do not produce the same labor-income response.
- **Main credibility result.** The instrument strongly predicts negative equity, but does not predict origination income, credit score, interest rate, Alt-A status, or no-documentation loans. The result also survives household fixed effects, very rich region-time controls, same-MSA-year comparisons, and income-hiding checks.
- **Mechanism message.** The paper cannot perfectly separate mechanisms, but the nonlinear pattern is most consistent with housing lock and/or financial distress. A pure wealth-effect story is less natural because positive home-equity gains do not generate mirror-image labor-supply increases.
- **Economic takeaway.** Household balance-sheet distress matters for labor supply. This gives a micro channel through which housing busts can deepen macro downturns.

2 Data and measurement (key objects)

2.1 Main data environment

- The main data come from a large U.S. financial institution, denoted *MyBank*, with transaction-level information from 2010 to 2014.

- The institution covers more than 31 million U.S. households over 2009 to 2013 and has borrowers in all 50 states.
- The paper links three broad data sources within the institution:
 - retail-account transactions, used to measure household income from deposit inflows;
 - mortgage-account records, used to observe origination terms and current loan-to-value measures;
 - credit-card / credit-bureau data, used to observe liabilities across lenders and updated credit scores.

Interpretation. The crucial advantage of the data is that the paper observes *actual* deposit inflows and detailed liabilities at monthly frequency, rather than relying on self-reported survey answers.

2.2 Income measurement from deposits

Retail accounts include checking and savings accounts. The paper treats deposit inflows as a high-frequency proxy for labor income.

The raw transaction data include the day, amount, and broad transaction type. The analysis focuses on transactions labeled as deposits, such as:

- direct deposits,
- teller and ATM deposits,
- and other deposit types such as mobile remote deposit capture.

To avoid accounts that are not meaningful for labor-income measurement, the author restricts to households with active retail accounts that appear to receive the majority of income.

A household is “active” if, in a given month, it deposits at least \$100 or holds at least \$200 in financial assets. To enter the panel, households must have at least 12 months of deposits in a moderate range and mean and median deposits above \$500.

Interpretation. The income measure is not wages from payroll files. It is broader household deposit income. But because the data are administrative and high-frequency, they are much less noisy than household surveys for this question.

2.3 Validation of the income measure

The paper validates deposit-based income in two ways.

- **Zip-code validation against IRS data.** Average annual income inferred from deposits is compared to IRS Statistics of Income zip-code data for 2010 to 2013. Correlations range from 0.736 to 0.911.
- **Origination-income validation.** For matched mortgage borrowers, income at origination and later deposit-based income have similar distributions. Household-level correlations range from 0.378 to 0.449 depending on the deposit measure.

The paper also isolates social-security/disability-type deposits using recurring payment timing rules and finds an average payment of about \$1,268 per recipient, very close to the Social Security Administration benchmark of about \$1,223.

Interpretation. These checks are important because the dependent variable is central to the paper. The author spends real effort showing that deposit flows are a sensible proxy for household labor income and benefit income.

2.4 Mortgage, liabilities, and leverage variables

Mortgage data include:

- mortgage type,
- origination date,
- reported income at origination,
- credit score at origination,
- interest rate at origination,
- origination LTV,
- current LTV based on internal house-price updates,
- delinquency and loss-mitigation information.

Credit-bureau data include liabilities across all lenders: mortgages, auto loans, student loans, HE-LOCs, credit cards, installment credit, and monthly credit scores.

2.5 Sample construction

- Main mortgage-plus-retail sample: about 200,000 households and 7.8 million household-month observations.
- Main analysis sample after requiring origination income, origination date, and additional information: about 5.4 million household-month observations.
- Robustness sample with retail and credit-card accounts and mortgages from any lender: about 20.1 million household-month observations.

2.6 Summary statistics (Table I)

For households with both retail accounts and a MyBank mortgage:

- mean monthly income (all deposits): \$7,663; median: \$5,315;
- mean monthly income from filtered direct deposits: \$6,470; median: \$5,172;
- mean savings: \$35,370; median: \$10,100;
- mean total liabilities in the credit-bureau subsample: \$266,300; median: \$225,000;

- mean FICO-like bank credit score: 767; median: 782;
- mean primary mortgage balance: \$199,900; median: \$170,700;
- mean mortgage interest rate at origination: 5.373%;
- mean mortgage age: 64 months;
- mean income at origination: \$7,494; median: \$6,237;
- mean origination LTV: 64%; mean current LTV: 58%;
- 92.0% of properties are owner occupied.

Interpretation. The sample looks like a broad middle-class mortgagor sample rather than an extreme distressed-borrower sample. That matters for external validity.

2.7 Dependent variables

The paper studies household labor supply through deposit income in two main forms:

1. **Dollar deposits per month** (raw monthly deposit inflows), denoted informally as \$Dep.
2. **Percent change in deposits relative to origination income**, denoted informally as %ΔDep, where the numerator is current monthly deposit inflow and the denominator is income reported at mortgage origination.

Interpretation. The dollar specification gives economic magnitude in levels. The normalized specification makes households more comparable across the income distribution.

2.8 Current LTV, negative equity, and synthetic LTV

The paper distinguishes three related objects:

- **Actual LTV:** current loan balance divided by current house value.
- **Negative home equity / underwater status:** an indicator for whether current LTV exceeds 100%.
- **Synthetic LTV (SLTV):** an instrument-like measure of current leverage that strips out household-specific balance choices and uses only timing of purchase, a synthetic amortization path, and local house-price changes.

3 Models and empirical specifications (all equations + interpretation)

3.1 Intuition: a flexible reduced-form around LTV buckets

The paper first studies whether income declines appear only at high LTV levels:

$$y_{icrt} = \alpha + \gamma_{rt} + \phi_{ct} + \sum_k \delta_{1k} \cdot \mathbf{1}\{l_k \leq LTV_{it} \leq h_k\} + \varepsilon_{icrt}. \quad (1)$$

Here:

- i indexes household,
- t month,
- r region,
- c mortgage origination cohort,
- y_{icrt} is household income,
- γ_{rt} are region-by-time fixed effects,
- ϕ_{ct} are origination-date-by-time fixed effects.

Interpretation. This is a diagnostic specification, not the final causal design. It asks whether income declines are concentrated only once leverage gets close to negative equity, which would be consistent with mechanisms like housing lock or distress rather than smooth wealth effects.

3.2 Decomposing current leverage

Actual current LTV can be written as

$$LTV_{it} \equiv LTV_{ic} \times \frac{1 + \% \Delta Loan_{it}}{1 + \% \Delta H P_{rct}}. \quad (2)$$

This shows that current leverage depends on:

- origination leverage LTV_{ic} ,
- how much the loan balance has changed since origination,
- how much local house prices have changed since origination.

Interpretation. This decomposition makes clear why raw LTV is endogenous. Prepayment and amortization can depend on household income, and origination LTV can depend on borrower choices and lender screening.

3.3 Synthetic loan path

To eliminate household-specific balance changes from the instrument, the paper replaces actual loan amortization with the synthetic balance path of a 30-year fixed-rate mortgage at the sample median mortgage rate:

$$\% \Delta SynthLoan_{ct} \equiv -\frac{(1+r)^{t-c} - 1}{(1+r)^T - 1}, \quad (3)$$

where $T = 360$ months and r is the median national monthly mortgage rate used by the author.

Interpretation. The synthetic balance change depends only on loan age, not on household behavior. This is how the design removes the possibility that richer households mechanically drive down their LTV by prepaying more.

3.4 Synthetic LTV instrument

The paper then constructs

$$SLTV_{rct} \equiv LTV_c \times \frac{1 + \% \Delta SynthLoan_{ct}}{1 + \% \Delta HPI_{rct}}, \quad (4)$$

where LTV_c is the median national origination LTV for cohort c and $\% \Delta HPI_{rct}$ is local house-price growth since origination.

The instrument used in the baseline 2SLS is the indicator

$$\mathbf{1}\{SLTV_{rct} \geq 100\}. \quad (5)$$

Interpretation. SLTV varies at the region-time-cohort level. Conditional on region-time and cohort-time fixed effects, the identifying variation comes from whether a household bought at a relatively fortunate or unfortunate time in its local market.

3.5 Two-stage least squares specification

Let underwater status be

$$U_{it} \equiv \mathbf{1}\{LTV_{it} \geq 100\}. \quad (6)$$

The paper estimates

$$U_{it} = \alpha + \gamma_{rt} + \phi_{ct} + \delta_1 \cdot \mathbf{1}\{SLTV_{rct} \geq 100\} + X'_{it}\beta + \eta_{icrt}, \quad (7)$$

$$y_{icrt} = \alpha + \gamma_{rt} + \phi_{ct} + \delta_2 \cdot \hat{U}_{it} + X'_{it}\beta + \varepsilon_{icrt}. \quad (8)$$

Interpretation. The first stage asks whether synthetic negative equity predicts actual negative equity. The second stage asks whether the part of underwater status driven by purchase timing and local house prices lowers labor income.

3.6 Core fixed effects and why they matter

The main specifications layer in several fixed effects:

- **MSA $\times$ time fixed effects:** absorb local labor-demand shocks and local housing-market shocks common within an MSA-month.
- **Origination date $\times$ time fixed effects:** absorb national cohort effects and changes in lending standards by origination cohort.
- **Household fixed effects:** absorb time-invariant borrower heterogeneity.

Interpretation. The identifying comparison is very demanding: households who bought at different times but live through the same later local shock and belong to the same lender-era environment.

3.7 Nonlinear and placebo specifications

The paper also studies nonlinear reduced forms using bins of actual LTV, house-price appreciation since origination, and SLTV. It also reruns the reduced form on the subsample with *positive* home equity only.

Interpretation. These nonlinear plots are central. They show that the labor-income response is concentrated around negative equity, rather than appearing smoothly for all house-price declines.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the effect

There is one formal instrument: whether synthetic LTV exceeds 100%. The credibility of the design comes from the fact that synthetic LTV depends only on:

- when the mortgage was originated,
- the cohort-level median LTV at origination,
- a synthetic amortization path,
- and MSA-level house-price changes.

It does *not* depend on household-specific prepayment, refinancing, home improvement, or active balance choices.

4.2 Relevance: the first stage is extremely strong

Table II, column (1), shows that an $SLTV > 1$ indicator raises the probability of actual negative equity by 0.623, or 62.3 percentage points.

Interpretation. Relevance is not a close call. The first stage is mechanically and economically very strong because purchase timing matters a lot for whether households end up underwater after a housing bust.

4.3 Why reverse causality is weak

A core reverse-causality concern is that weak-income households may choose different mortgages, prepay differently, or invest differently in home maintenance. The design weakens these concerns in two ways:

- it replaces actual balance change with a synthetic amortization path;
- it uses local house-price changes and purchase timing rather than household behavior to generate identifying variation.

4.4 Placebo tests at origination

If the instrument were proxying for borrower quality or local labor sensitivity, it should predict characteristics known at origination.

It does not.

Table II shows that $SLTV > 1$ does *not* significantly predict:

- reported monthly income at origination,
- credit score at origination,
- mortgage interest rate at origination.

Table III adds harder-to-fake checks and shows no meaningful relation with:

- Alt-A status,
- no-income-documentation status,
- verified income at origination.

Interpretation. This is powerful because it goes directly at the concern that the lender happened to expand riskier lending in certain regions at certain times.

4.5 Household fixed effects and within-household variation

When household fixed effects are added, the reduced form remains negative and significant. In Table III, column (4), the reduced-form coefficient on $SLTV > 1$ is about -1.37 percentage points for normalized deposit income.

Interpretation. This means the effect is not just that different *types* of households bought at different times. The same household is observed over time, and income falls when the household is pushed into negative equity by the identified source of variation.

4.6 Positive-equity placebo

A particularly strong placebo is to remove all household-months in which households actually have negative equity and re-estimate the reduced form.

In Table III, columns (5) and (6), the relationship disappears. The coefficient on $SLTV > 1$ is about 0.08 and statistically insignificant, and the interaction with local MSA deposit growth is also insignificant.

Interpretation. This is one of the cleanest pieces of identification in the paper. The same local house-price timing variation is not associated with lower income unless it actually produces negative equity.

4.7 Within-MSA, same-year, and labor-market timing checks

The paper strengthens the design in two further ways:

- **Zip $\times$ time fixed effects (Table IV, column 3).** The result remains around -3.77% , suggesting within-MSA selection is not driving the result.
- **MSA $\times$ time $\times$ origination-year fixed effects (Table IV, column 5).** Comparing households who bought in the same MSA and year but only a few months apart still yields a significant effect of about -2.20% .
- **MSA $\times$ time $\times$ graduation-year fixed effects in the student-loan subsample (Table IV, column 6).** Controlling for approximate labor-market entry timing still yields an effect of about -5.63% .

Interpretation. These checks make it increasingly hard to say the result is really about long-run local labor-market composition or borrower age/cohort differences.

4.8 Income-hiding concern

Because the income measure comes from deposits at the same institution that often owns or services the mortgage, borrowers might try to shift deposits away to look poorer.

The paper addresses this several ways:

- it restricts to active deposit accounts;
- it shows the result is not driven by zero-deposit cases;
- it shows the effect comes from large income drops rather than small systematic drifts;
- it finds households become *more* likely to receive Social Security or disability-type checks;
- it reruns reduced forms on households whose mortgages are at another lender or not serviced by MyBank, where there is no reason to hide income from MyBank.

Interpretation. The deposit-shifting objection is taken very seriously, and the evidence goes against it.

4.9 What the paper can and cannot distinguish

The paper convincingly identifies a causal effect of negative home equity on labor income, but it does *not* fully separate the underlying mechanisms. The evidence is most consistent with housing lock and/or financial distress. Entrepreneurship collateral effects or standard wealth effects are less consistent with the nonlinear patterns, though they cannot be ruled out entirely.

5 Tables and figures

5.1 Table I: summary statistics

Read: The table establishes what the sample looks like: a broad mortgage-borrower population with median monthly income around \$5,000 to \$6,000, median primary mortgage balances around \$171,000, median credit scores around 782, and mean current LTV around 58%.

Economic message. The paper is not about a tiny tail of already-distressed borrowers. It studies a large, fairly representative mortgagor sample in which negative equity becomes economically meaningful because of the housing bust.

Summary Statistics				
This table presents summary statistics for MyBank data. To be included in the panel, households must have at least 12 months with deposits across all accounts >\$300 and <\$80,000 and a mean and median level of deposits across all accounts >\$500 and <\$25,000. For direct deposits, households must have at least 12 months of direct deposits >\$100 and <\$25,000, a mean and median level of direct deposits across all accounts >\$500 and <\$25,000, and >75% of all deposits must be via the direct deposit channel. All data are winsorized at 99 th percentile. This sample includes only those households that have retail and mortgage accounts at MyBank over the period 2010 to 2014.				
	Mean	Median	SD	Number of Obs (Millions)
Households w/ MyBank Retail Accounts and MyBank Mortgage 2010 to 2014				
Retail data				
Income (all)	\$7,663	\$5,315	\$8,459	7.835
Income (dir. dep.)	\$4,142	\$2,826	\$4,742	7.835
Income (dir. dep. w/ filter)	\$6,470	\$5,172	\$5,226	2.201
Savings	\$35,370	\$10,100	\$60,626	7.835
Card/Credit Bureau Data (w/ MyBank credit card account)				
All liabilities	\$365,300	\$225,000	\$210,610	5.158
Has auto loan	30%			5.158
But used/available all credit	29%	19%	29.3%	5.158
FCO bank credit score	767	752	74.4	5.158
Mortgage data				
Primary MTD balance	\$199,900	\$170,700	\$137,130	7.835
MTD interest rate @ origination	5.373	5.375	1.227	7.835
MTD age (months)	64	58	49	7.835
Income @ origination	\$7,494	\$6,237	\$5,171	5.419
Origination loan-to-value (%)	64	68	22.1	7.835
Current loan-to-value (%)	58	58	31.5	7.835
Is owner occupied	92.0%			7.835
Subprime	10.2%			7.835
Jumbo	19.3%			7.835
Stated income	13.5%			7.835
Single-family residential	88.7%			7.835
Is fixed rate	83.9%			7.835

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Table II Validity of SLTV Instrument and Observables					
This table provides evidence that after controlling for region-time and origination date-time fixed effects, a household's synthetic loan-to-value ratio (SLTV) greater than 100% is a valid instrument to examine the effect of negative home equity on household labor supply based on observable characteristics. SLTV is an instrument for LTV that does not depend on household-specific factors, except the timing/location of a move, and that varies at the region-time-cohort level. Column (1) regresses a dummy equal to 1 if a household's current LTV is greater than 100% on a dummy equal to 1 if the household's SLTV is greater than 100%, after including MSA × time, and origination date × time fixed effects. This gives the first stage estimate of an IV regression. Column (2) is the same as column (1), but uses the percent change in deposits as the dependent variable, where the numerator is the monthly deposit inflows and the denominator is the household's income at the time of mortgage origination. Column (3) is the same as column (1), but uses raw monthly deposits as the dependent variable, without any normalization. Column (4) is the same as column (1), but uses monthly gross reported income at origination as the dependent variable. Column (5) is the same as column (1), but uses the credit score at origination as the dependent variable. Column (6) is the same as column (1), but uses the initial mortgage interest rate at origination as the dependent variable. All standard errors are clustered at the MSA level. p-Values: *10%, **5%, and ***1%.					
	First Stage	Reduced Form		@Origination Placebo Tests	
	(1)	(2)	(3)	(4)	(5)
	LTV > 1	%ΔDep	\$Dep	\$ Mo. Income	Credit Score
SLTV > 1	0.623*** (0.028)	-4.42*** (0.775)	-436.8*** (128.5)	-64.8 (151.9)	1.03 (1.66)
MSA × Time FE	Y	Y	Y	Y	Y
Orig. Date × Time FE	Y	Y	Y	Y	Y
Observations (mil)	5.375	5.375	5.375	5.375	5.375

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5.2 Table II: validity of the SLTV instrument and observables

Read: Column (1) gives the first stage: $SLTV > 1$ raises actual underwater status by 62.3 percentage points. Columns (2) and (3) show reduced-form declines in income: about -4.42% in normalized deposits and about $-\$436.8$ in dollar deposits. Columns (4) to (6) show no meaningful relation with origination income, credit score, or interest rate.

Economic message. This is the paper's first major credibility table. The instrument is strong, and the same timing-based variation that predicts negative equity does *not* look like a proxy for ex ante borrower quality.

5.3 Table III: robustness to selection on unobservables

Table III Robust to Selection on Unobservables						
This table provides evidence that after controlling for region-time, origination date-time, and household fixed effects, a household's synthetic loan-to-value ratio (SLTV) greater than 100% is a valid instrument to examine the effect of negative home equity on household labor supply, focusing on tests that reveal differences in difficult-to-observe or unobservable characteristics. SLTV is an instrument for LTV that does not depend on household-specific factors, except the timing/location of a move and that varies at the region-time-cohort level. Column (1) regresses a dummy equal to 1 if a household's mortgage at origination was "Alt-A" or a "Liar Loan" on a dummy equal to 1 if the household's SLTV is greater than 100%, after including MSA × time and origination date × time fixed effects. Column (2) is the same as column (1), but uses a dummy variable equal to 1 if the mortgage has no documentation at origination as the dependent variable. Column (3) is the same as column (1), but uses monthly gross verified income at origination as the dependent variable. Column (4) is the same as column (1), but uses the percent change in deposits as the dependent variable, where the numerator is monthly deposit inflows and the denominator is the household's income at the time of mortgage origination and the regression includes household fixed effects. Column (5) is the same as column (4), but excludes observations for which a household has negative home equity. Column (6) is the same as column (5), but includes the interaction between negative SLTV and demeaned monthly percent changes in aggregate deposits by MSA. All standard errors are clustered at the MSA level. p-Values: *10%, ** 5%, and ***1%.						
	@Origination Placebo Tests			HH FEs		
	(1)	(2)	(3)	(4)	(5)	(6)
	Alt-A	No Income Docs	Verified 8 Mo. Income	%ΔDep	%ΔDep	%ΔDep
SLTV > 1	0.0002 (0.0037)	-0.004 (0.013)	-6.1 (122.5)	-1.37*** (0.42)	0.08 (0.61)	0.08 (0.61)
SLTV > 1 × %MSA Dep					0.17 (0.24)	
MSA × Time FE	Y	Y	Y	Y	Y	Y
Orig. Date × Time FE	Y	Y	Y	Y	Y	Y
HH FE	N	N	N	Y	Y	Y
Sample	All	All	All	All	Eq > 0	Eq > 0
Adjusted R ²	0.110	0.107	0.056	0.480	0.529	0.531
Observations (mil)	5.375	5.375	4.144	5.375	4.763	4.763

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Read: Columns (1) to (3) show no significant relation of the instrument with Alt-A status, no-documentation loans, or verified income at origination. Column (4) shows a negative reduced form even with household fixed effects (-1.37). Columns (5) and (6) show that once the sample excludes all negative-equity observations, the reduced form disappears.

Economic message. This is the table that turns a strong correlation into a much more credible causal story. The effect is not about hidden borrower risk. It turns on only when the shock actually generates negative equity.

5.4 Figure 1 and Figure 2: the effect is nonlinear and tied to negative equity

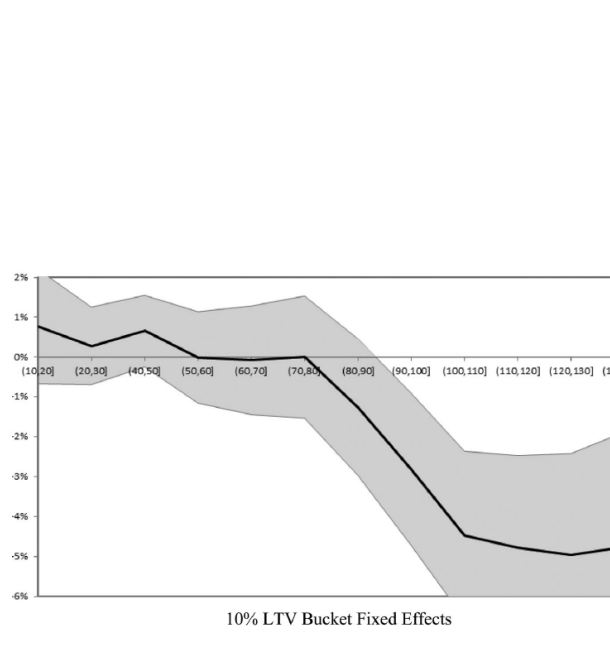


Figure 1: income by actual LTV bucket

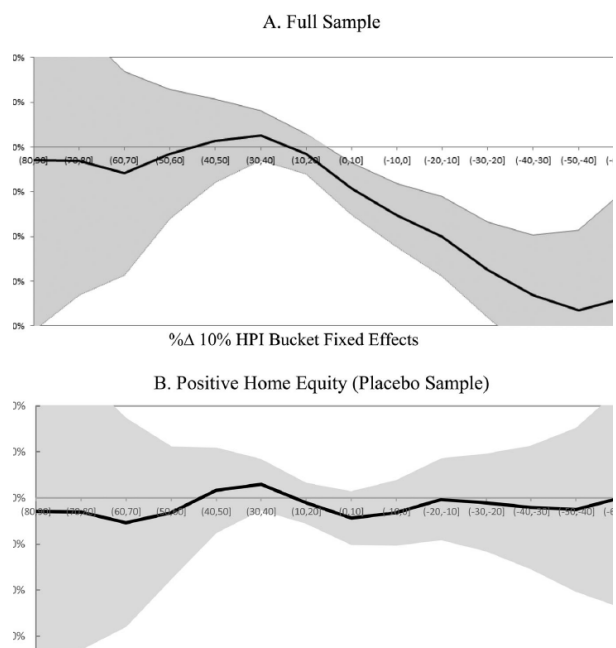


Figure 2: income by house-price-change bucket, with positive-equity placebo

Read: Figure 1 shows that income is roughly flat across low and moderate LTV buckets, then falls sharply once LTV approaches and exceeds 100%. Figure 2 shows a similar reduced-form pattern using house-price changes since origination; in the placebo sample restricted to positive-equity households, the decline largely disappears.

Economic message. The paper is not estimating a smooth “less housing wealth, less labor” slope. The action is around the negative-equity region. That pattern lines up better with debt-related frictions than with standard wealth effects.

5.5 Figure 3: first stage and reduced form line up at the 100% threshold

Read: The blue line shows that the probability of negative equity rises sharply when SLTV crosses 100%. The red line shows that income falls sharply in the same region. Away from that region, large movements in SLTV produce little income effect.

Economic message. This is perhaps the single most persuasive graph in the paper. It visually

links the instrument's first stage and the reduced form, and it shows both are highly nonlinear exactly where theory says they should be.

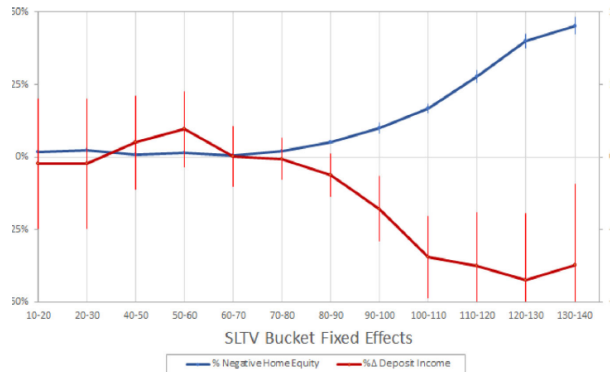


Figure 3. SLTV, negative home equity, and income. This figure shows the relationship

Table IV
Negative Home Equity and Labor Supply

This table shows the average change in household income associated with negative household home equity using variation in the timing of home purchase as an instrument for the probability of having negative equity. The instrument is a dummy variable equal to 1 if a household's synthetic loan-to-value ratio (SLTV) is greater than 100% after controlling for MSA, time, origination date, time, and household fixed effects. Column (1) reports the results of running the 2SLS regression using raw monthly deposit inflows as the dependent variable, without any normalization, on instrumented negative home equity. Column (2) is the same as column (1) but the dependent variable is the percent change in deposits, where the numerator is the monthly deposit inflows and the denominator is the household's income at the time of mortgage origination. Column (3) is the same as column (2), but includes zip-time instead of MSA-time fixed effects. Column (4) is the same as column (2), but includes time-varying nonparametric household-level controls. The controls include dummies for origination income and property value, original mortgage interest rate by percentage buckets, and original credit score in bins of 50 all interacted with time fixed effects. Column (5) is the same as column (2), but includes MSA $\times$ time $\times$ origination year fixed effects. Column (6) is the same as column (2), but instead of origination date $\times$ time fixed effects it includes graduation year $\times$ MSA $\times$ time fixed effects among the subset of borrowers with outstanding student loans. All standard errors are clustered at the MSA level. p -Values: *10%, **5%, and ***1%.

	(1) \$Dep	(2) %ΔDep	Zip FE (3) %ΔDep	X _{it} Controls (4) %ΔDep	Orig Yr $\times$ MSA FE (5) %ΔDep	Grad Yr FEs (6) %ΔDep
LTV > 1 (IV: SLTV > 1)	−298.1*** (61.3)	−3.47*** (1.18)	−3.77*** (1.13)	−4.94*** (1.03)	−2.20** (0.89)	−5.63*** (2.97)
Region $\times$ Time FE	YMSA	YMSA	YZIP	YMSA	N/A	YMSA
Orig. Date $\times$ Time FE	Y	Y	Y	Y	N/A	N
HH FE	Y	Y	Y	Y	Y	Y
HH time-varying controls	N	N	N	Y	N	N
MSA $\times$ Time $\times$ Orig Yr FE	N	N	N	N	Y	N
MSA $\times$ Time $\times$ Grad Yr FE	N	N	N	N	N	Y
F-stat	2,440.4	2,440.4	2,109.6	2,394.8	119.25	126.1
Adjusted R ²	0.380	0.490	0.520	0.492	0.620	0.550
Observations (millions)	5.375	5.375	5.271	5.219	5.219	0.665

5.6 Table IV: 2SLS estimates of negative home equity on labor supply

Read: The baseline 2SLS estimate is about $-\$298$ per month in dollar deposits and -3.47% in normalized deposit income. The estimate remains economically similar with zip-time fixed effects (-3.77%), rich household-time-varying controls (-4.94%), same-MSA-year comparisons (-2.20%), and graduation-year-by-MSA-time controls in the student-loan subsample (-5.63%).

Economic message. The magnitude is not only statistically significant; it is also robust across identification-tightening specifications. This is the table that establishes the headline causal effect.

5.7 Table V: robustness to income hiding

Table V
Robust to Income "Hiding"

This table explores the drivers of the negative effect of mortgage LTV on labor supply and shows that it is not driven by "hiding" deposits with other institutions. Just as in the main specifications, column (1) represents the percent change in deposits, where the numerator is the monthly deposit inflows and the denominator is the household's income at the time of mortgage origination, on an instrumented dummy equal to 1 if current mortgage LTV is greater than 100%, and MSA $\times$ time, origination date $\times$ time, and household fixed effects. A dummy equal to 1 if any synthetic loan-to-value ratio (SLTV) measure is greater than 100% is used as an instrument for the likelihood that a household has negative home equity. SLTV is an instrument for LTV that does not depend on household-specific factors except the time of a move and that varies at the region-time-cohort level. In this case, however, observations with a 100% decline in deposits are excluded from the analysis. Column (2) is the same as column (1) but excludes changes larger than 25%. Column (3) is the same as column (1), but does not exclude any deposits and the dependent variable is a dummy equal to 1 if the household receives social security checks, which are defined as direct deposits received on the third of the month, or the second, third, or fourth Wednesday that are not explained by regularly scheduled labor-related direct deposits. Column (4) restricts the sample to the subset of borrowers with credit cards and associated credit bureau data. For this subset, I do not rely on mortgage data, so I know the approximate timing and zip code of origination (see the text for more details), but not the actual LTV or income at origination. I therefore regress the dollar amount of deposits per month on the percentage of mortgages in the estimated zip code $\times$ origination year $\times$ time with negative home equity, after including MSA $\times$ time, origination date $\times$ time, and household fixed effects. Column (5) is the same as column (4), but restricts the analysis to households with mortgages not serviced or owned by *MyBank*. All standard errors are clustered at the MSA level. p -Values: *10%, **5%, and ***1%.

	(1) %ΔDep	(2) %ΔDep	(3) %GeSS	(4) \$Dep	(5) \$Dep
LTV > 1 (IV: SLTV > 1)	−3.35*** (1.12)	−0.07 (0.55)	0.91*** (0.32)		
%NegIn (Region $\times$ Cohort $\times$ Time)				−48.8*** (10.4)	−65.0*** (15.0)
Region $\times$ Time FE	Y	Y	Y	Y	Y
HH FE	Y	Y	Y	Y	Y
Orig. Date $\times$ Time FE	Y	Y	Y	Y	Y
Normalization	Orig Inc	Orig Inc	No	N/A	N/A
Debtors constraint	> 25%	> 25%	N/A	N/A	N/A
Mortgage servicer/borrower	All	All	All	All	Not MyBank
Orig location/date method	Actual	Actual	Actual	Derived	Derived
Adjusted R ²	0.623	0.430	0.548	0.344	0.348
Observations (mil)	4.794	3.888	5.375	20.113	15.018

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Read: Column (1) shows the result survives after excluding cases with zero deposits. Column (2) shows the effect is driven by large declines in income; once changes larger than 25% are excluded, the normalized effect disappears. Column (3) shows negative equity increases the probability of receiving Social Security or disability checks by about 0.91 percentage points. Columns (4) and

(5) show reduced-form negative deposit effects even when the mortgage is outside MyBank or not serviced by MyBank.

Economic message. This table argues against the interpretation that borrowers are merely hiding income from their lender. The pattern looks more like real labor-income declines, including movement toward retirement or disability-type income.

6 Short summary

- The paper asks whether negative home equity reduces labor supply.
- It uses administrative data on deposits, liabilities, and mortgages for a very large sample of U.S. households.
- The core instrument is a timing-based synthetic LTV measure built from local house-price changes and a synthetic amortization path.
- The first stage is strong: synthetic negative equity strongly predicts actual negative equity.
- Instrumented negative home equity lowers household labor income by roughly 2.2% to 5.6%, with a baseline estimate around 3.47%.
- The effect is highly nonlinear and appears only near or beyond the negative-equity threshold.
- Placebo tests at origination and in positive-equity samples support causal interpretation.
- The result survives household fixed effects, same-MSA-year comparisons, and income-hiding robustness tests.
- The most plausible mechanisms are housing lock and financial distress.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that negative home equity has a causal negative effect on household labor income. The contribution is not just that housing busts and labor-market weakness move together, but that the household balance-sheet channel itself matters for labor supply.

7.2 Economic intuition

When households fall into negative equity, housing debt becomes much more distortionary. Mobility can be constrained, default and distress become more salient, and the labor market can no longer be treated as separate from the household balance sheet. That is why the estimates are nonlinear: crossing into negative equity is not just “less wealth”; it changes the whole optimization problem facing the household.

7.3 Takeaway

This paper gives a clear microfoundation for why housing crashes can deepen recessions. A fall in house prices does not only depress demand through spending. By pushing households underwater, it can also reduce labor supply and slow the recovery of employment and income.